

Topic 4: Statistics

4.1 Data

Subject content	Guidance	Curriculum reference
A Find the mode and mean of a set of data presented in a list, table or bar chart	Find the mode or modal value	S7.1A S7.1C
B Find the median and range of a set of data presented in a list, table or bar chart	Odd or even number of values (so middle value, or halfway between two middle values)	S7.1B
C Calculate the mean from a frequency table using grouped and ungrouped data	$\frac{\sum fx}{\sum f}$	S8.1A S9.1G
D Calculate an estimate of the range and mean from grouped data		S9.1F
D Find the modal class of grouped data		S7.1D
E Compare two sets of data using their ranges, averages or the shape of the graph	Mohamed's mean race time is 54 seconds, and the range of his race times is 10 seconds. Umar's mean race time is 56, and the range of his race times is 4 seconds. Who is the better runner? You must explain your answer.	S7.1E S8.1C
F Identify and suggest reasons for outliers in data	Which value is an outlier? Suggest a reason for this.	S9.1C S9.1H
G Choose the most appropriate average to use	Should you use mean, median or mode to compare the average size of shoes sold in 2 shoe shops? Explain your answer.	S8.1B
H Solve problems involving mean, median, mode, and range involving comparing data	Five numbers have a mean of 8 Four of the numbers are: 9, 7, 9 and 11 What is the fifth number?	S7.1G S8.1D

4.1 Data *continued*

Subject content	Guidance	Curriculum reference
I Design and use data-collection sheets and tables	A student wants to record the gender and hair colour of everyone in his class. Design a data-collection sheet that he could use to record their responses.	S9.1E
J Know and use correct set language and notation		S9.1J
K Choose a suitable sample size and what data to collect		S9.1B